



The Loop Hypothesis

Artist's Proof 04

Cyclic Cosmology

*A structural identity between black-hole interiors and the origin of expanding
universes*

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0 — Dependency and Scope

Artist's Note

This is Artist's Proof 04 of The 420 Code, a chapter of Notebook VI — Cosmology. It proposes a structural identity: that a black-hole interior at the quantum-gravity scale and the origin of an expanding universe are the same structure, read from two sides of a transition hypersurface. The inside of a black hole is a collapsing universe; the maximal extension reveals an expanding universe on the other side. The mathematics of this identity — the Schwarzschild interior as a Kantowski-Sachs cosmology, the Kruskal-Szekeres extension — has been published since 1966. AP04 reads it as atemporal: not a bounce in time, but one structure foliated two ways.

Because this paper reaches across established physics, speculative-but-published physics, and philosophical interpretation, it carries a three-layer certainty system, and every claim is tagged with its layer. Layer 1 is established (the diffeomorphism, the entropy bounds, the Page curve). Layer 2 is speculative but published (Popławski's torsion mechanism for singularity avoidance and isotropisation). Layer 3 is philosophical interpretation (the structural identity itself, the awareness identification, the ethics, the formalisation conjecture). The central result is Conjecture L7 — the metric equation $g^+ = (\varphi^{-1})^*(g^-) + \varepsilon \cdot h$ — which formalises the axiom 1:1 + $1 \times \varepsilon$ across the transition under five explicit assumptions. It is a conjecture, not a derivation; the paper carries nine kill switches, four of which — KS-L0.1, KS-L0.2, KS-L0.8, KS-L0.9 — guard the conjecture directly.

The sharpest obstacle is named openly: isotropising the anisotropic Kantowski-Sachs interior into the isotropic FRW cosmology we observe (KS-L0.8). The fine-structure constant is acknowledged as not derived (D3); its underivability is consistent with the structural argument in AP41 (you cannot read the address from inside the room). The paper does not protect itself — it tells you where to cut. Nothing is imported beyond the cited established and published physics; the axiom and its consequences descend from {S, B, R, C}.

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Orientation

Where this paper fits in the 420 Code

AP04 is a chapter of Notebook VI (Cosmology) — the loop hypothesis itself. It is the conjectural step that AP41 (The Loop) inherits at Stage 6 (the door): AP41's KS-41.3 inherits AP04's KS-L0.1. AP04's two constraints (the compression constraint ρ_p and the propagation constraint c) are the same constants whose conjugacy AP03 conjectures; if AP03's conjugacy holds, the two constraints are not independent. The paper draws on established relativity (Kantowski & Sachs 1966; Kruskal-Szekeres) and published Einstein-Cartan torsion work (Popławski).

How to read this paper

Read the layer tags. Layer 1 claims are established and carry citations; Layer 2 claims are published but not confirmed; Layer 3 claims are philosophical and not testable in the same way. The formal claims L1-L10, the interpretations I1-I2, the Conjecture L7, and the predictions P1-P6 are preserved verbatim and close with the box mark ■. The conjecture is the spine; the nine kill switches (section 13) are where it can be cut.

0.1 — What this paper does

It proposes a structural identity between black-hole interiors at the quantum-gravity scale and the origin of expanding universes, and formalises the axiom $1:1 + 1 \times \varepsilon$ as a metric equation across a transition hypersurface: Conjecture L7, $g^+ = (\varphi^{-1})^*(g^-) + \varepsilon \cdot h$, under five explicit assumptions. It introduces a three-layer certainty system (established / speculative-but-published / philosophical) and tags every claim. It identifies the two fundamental constraints (c and ρ_p), reads the symmetry break through the electron and α , identifies awareness with $\varepsilon > 0$, and lays out nine falsification conditions. It is a conjecture, not a derivation.

0.2 — Dependencies

Established relativity: the Schwarzschild interior as Kantowski-Sachs (Kantowski & Sachs 1966), the Kruskal-Szekeres maximal extension, the

Bekenstein-Hawking entropy bound, the Page curve and island formula (Almheiri et al. 2021). Published speculative physics: Popławski's Einstein-Cartan torsion mechanism (2010, 2016, 2023). Within the 420 Code: AP03 (the conjugacy of c and ρ_p). The Embedding Hypothesis (EH) and Quantum-Record Alignment (QRA) are proved in AP20; the conjecture itself is held at Layer 2/3 and is not claimed as proved.

0.3 — Axiom mapping

Axiom S (Symmetry — the involution σ). 1:1 is interpreted as a $\mathbb{Z}_2$ symmetry on the transition hypersurface Σ (Interpretation I1): the transition diffeomorphism φ maps the collapsing geometry to the expanding one, $\varphi|_{\Sigma} = \text{id}$, normal-reversing.

Axiom B (Break — the unique break ε). $+1 \times \varepsilon$ is interpreted as the $\mathbb{Z}_2$ symmetry breaking on Σ (Interpretation I2). The perturbation $\varepsilon \cdot h$ is the single break carried across the transition — the records, the anisotropy difference, the initial conditions of expansion.

Axiom R (Record — monotone accumulation). Information is deviation from symmetry: if $\varepsilon = 0$ there are no records; if $\varepsilon > 0$, h carries the records compressed into the transition. The transition hypersurface is the point of maximum coarse-grained entropy.

Axiom C (Constraint — locality; finite propagation speed c). The propagation constraint c defines the null-cone structure; the compression constraint ρ_p is the Planck density where quantum gravity dominates. These are the two constraints between which the break is held.

0.4 — Epistemic status per section

1-3 (Method, axiom, layers). FOUNDATIONAL.

4-5 (Constraints, the break). LAYER 1 + LAYER 3 — established physics with architectural interpretation.

6 (Awareness). LAYER 3 — entirely philosophical; the awareness identification.

7 (Formalisation). LAYER 2 + 3 — Conjecture L7 under five explicit assumptions.

8 (The structural identity). LAYER 1 + 2 — established mathematics (the diffeomorphism, the entropy resolution) plus the speculative isotropisation mechanism.

9-12 (LQC, predictions, scale, existing physics). LAYER 1-3 — the relation to other programmes and the unique predictions.

0.5 — Kill switch summary

Nine kill switches, KS-L0.1 through KS-L0.9 (canonical registry numbering). KS-L0.1 (structural identity disproved) and KS-L0.2 (no singularity resolution) are LIVE — HARD and target the conjecture; KS-L0.8 (anisotropy irreconcilable) is the sharpest obstacle, LIVE — HARD; KS-L0.7 (fermions unnecessary) is LIVE — HARD; KS-L0.3 (information loss), KS-L0.4 (proton decay), KS-L0.5 (GSL violated), KS-L0.6 (actualisation without awareness / GRW), and KS-L0.9 (CMB mismatch) are LIVE — EMPIRICAL.

0.6 — Structural debts owed

D1 (Isotropisation). A rigorous mechanism for Kantowski-Sachs → FRW isotropisation — the sharpest obstacle (KS-L0.8). Popławski's torsion mechanism is proposed but not established. Open.

D2 (CMB spectrum). A derivation of the curvature perturbation spectrum $P_m(k)$ from $\varepsilon \cdot h$ — required for KS-L0.9. Open.

D3 (Fine-structure constant). A derivation of α 's value from the axiom — acknowledged as open. Its underivability is consistent with the structural argument in AP41 (α cannot be read from within the splinter; structurally undetermined at sub-leakage precision). Open.

0.7 — Items held open without claiming debt status

The awareness identification (section 6) is a Layer 3 naming choice with philosophical consequences; the physics is identical under either name, and it is falsifiable only via KS-L0.6 (GRW). The scale question (section 11) — whether

the identity is cosmic, astrophysical, or nested-fractal — is a Layer 3 reading. The interpretive identification of the electron and α (Step 6 of the chain) is physically motivated but not derived from the conjecture.

0.8 — Structural relationships

AP03 (The Ratio). The conjugacy of the compression constraint ρ_p and the propagation constraint c . If AP03 holds, AP04's two constraints are not independent.

AP41 (The Loop). Inherits AP04's conjecture at Stage 6 (the door): AP41's KS-41.3 inherits KS-L0.1. AP41's structural argument for the underivability of α is the body of work-level statement of AP04's D3.

AP20 (The Proof). EH and QRA proved; the substrate-as-manifold identity that the structural identity rests on at Layer 3.

Established relativity (Kantowski-Sachs, Kruskal-Szekeres). The Layer 1 mathematics: the black-hole interior as a collapsing cosmology, the maximal extension as an expanding one.

Popławski (Einstein-Cartan torsion). The Layer 2 mechanism for singularity avoidance and isotropisation — published, not confirmed; the load-bearing speculative input.

1 — On Method

This paper reaches across three kinds of certainty, and it does not pretend they are one. A three-layer certainty system is introduced: established physics, speculative-but-published physics, and philosophical interpretation. Every claim is tagged with its layer, so that a reader can see exactly how much weight each step carries. The argument does not protect itself; it tells you where to cut.

2 — The Axiom

Axiom L0. $1:1 + 1 \times \varepsilon$. One-to-one symmetry, broken by an infinitesimal perturbation $\varepsilon > 0$. The axiom is a philosophical choice, not a physical claim. It cannot be found inside the structure it generates.

3 — The Three Layers

Layer 1: Established physics. The diffeomorphism between the Schwarzschild interior and the Kantowski-Sachs cosmological model. The causal structure of the Kruskal-Szekeres maximal extension. The Bekenstein-Hawking entropy bound. The Page curve and island formula.

Layer 2: Speculative but published physics. Popławski's Einstein-Cartan torsion mechanism for singularity avoidance and isotropisation. Published in peer-reviewed journals but not experimentally confirmed and not mainstream.

Layer 3: Philosophical interpretation. The structural identity itself (atemporal, not a bounce). The awareness identification. The ethics derivation. The formalisation conjecture.

4 — The Two Constraints

4.1 — Propagation constraint: c

Claim L1 (Established). $c = 299,792,458$ m/s defines the null-cone structure of spacetime. Photons follow null geodesics ($ds^2 = 0$) with $d\tau = 0$. No inertial rest frame for massless particles. [Layer 1]

4.2 — Compression constraint: ρ_p

Claim L2 (Established). The Planck density $\rho_p = c^5/(\hbar G^2) \approx 5.16 \times 10^{96}$ kg/m³ is the natural density scale where quantum gravity is expected to dominate. [Layer 1]

4.3 — Inverse relationship

Claim L3 (Dimensional fact + interpretation). Planck units show c , G , $\hbar$ co-define the quantum-gravity regime. $\ell_p/t_p = c$ demonstrates dimensional coupling. Necessary but not sufficient for ontological equivalence. [Layer 1: dimensional fact. Layer 3: interpretation.]

5 — The Symmetry Break (ϵ)

5.1 — Primordial entropy

The symmetry break is read through the low entropy of the early universe: the initial state is special, and that specialness is the break expressed as the arrow of time. [Layer 1: the low-entropy initial condition. Layer 3: its identification with ϵ .]

5.2 — The electroweak bridge

The electroweak symmetry breaking of the Standard Model is the established physical instance of a symmetry that holds at high energy and breaks at low energy — the structural template for 1:1 + $1 \times \epsilon$ read at the particle scale. [Layer 1: established. Layer 3: architectural reading.]

5.3 — The electron

Observation L4 (Calculation). The electron violates the Kerr-Newman horizon condition ($a^2 + Q^2 \leq M^2$ in geometric units) by $\sim 10^{89}$ — the spin term alone exceeds extremality by $\sim 8 \times 10^{88}$ and the charge term by $\sim 4 \times 10^{42}$ — the maximal inverse of the compression constraint among known stable particles. [Layer 1: calculation. Layer 3: interpretation.]

5.4 — Fine-structure constant

Observation L5 (Established value + architectural interpretation). $\alpha = e^2/(4\pi\epsilon_0\hbar c) \approx 1/137$. Dimensionless coupling strength of the electromagnetic interaction. Architectural interpretation: the ratio of break strength (e^2) to constraint strength ($\hbar c$). Its value is not derived from the axiom (D3); the underivability is consistent with AP41's structural argument that α cannot be read from within the broken system. [Layer 1: established value. Layer 3: interpretation.]

6 — Awareness as Structural Minimum

Entirely Layer 3. Structurally necessary to the ethics derivation but not currently testable. Physicists may proceed to section 7.

Claim L6 (Philosophical, Layer 3, KS-L0.6). Awareness $\equiv \varepsilon > 0$. The actualisation state is self-referential. A naming choice with philosophical consequences. The physics is identical under either name.

Hierarchy. awareness ($\varepsilon > 0$) $\rightarrow$ actualisation (record formation) $\rightarrow$ consciousness (awareness aware of itself; contingent, not necessary).

Ethics (two-part result). First (structural): the base condition (awareness, $\varepsilon > 0$) is unitary — all actualisation shares a common origin in the symmetry break. A structural fact within the architecture. Second (perspectival): a conscious entity recognising this unity can identify cruelty as incoherent with it. The obligation is in the recognition, not in the structure.

Falsification. If objective collapse (GRW) is confirmed — states become actual without relational structure — the identification of ε with awareness is dead [Ghirardi et al. 1986] (KS-L0.6).

7 — The Formalisation: From Axiom to Conjecture

7.1 — Assumptions

Assumption A1. Working model: Einstein–Cartan gravity with fermionic matter and spin–torsion coupling.

Assumption A2. There exists a non-singular transition hypersurface Σ , defined by an invariant criterion (e.g. a curvature-scalar threshold or an effective quantum-gravity completion condition reaching the Planck regime), replacing the classical singularity.

Assumption A3. There exists an approximate $\mathbb{Z}_2$ mapping across Σ in the chosen completion.

Assumption A4. The expanding side is approximately FRW after Popławski’s torsion-driven isotropisation mechanism.

Assumption A5. Deviations between the collapsing and expanding geometries are captured by a perturbation of the form $\varepsilon \cdot h$, where ε is small and h is a symmetric 2-tensor on M .

7.2 — Interpreting 1:1

Interpretation I1. $1:1 \equiv \mathbb{Z}_2$ symmetry on Σ (Assumption A3).

7.3 — Interpreting $+ 1 \times \varepsilon$

Interpretation I2. $+ 1 \times \varepsilon \equiv \mathbb{Z}_2$ symmetry breaking on Σ .

7.4 — The conjecture

Conjecture L7 (The Loop Hypothesis, guarded by KS-L0.1, KS-L0.2, KS-L0.8, KS-L0.9). $g^+ = (\varphi^{-1})^*(g^-) + \varepsilon \cdot h$, where $\varphi: M^- \rightarrow M^+$ is the transition diffeomorphism ($\varphi|_\Sigma = \text{id}$, normal-reversing), $(\varphi^{-1})^*$ denotes the pullback of φ^{-1} (mapping tensors on M^- to tensors on M^+), ε is the symmetry-breaking

parameter, and h is the perturbation tensor encoding the anisotropy difference, the information content, and the initial conditions of expansion. [Layer 2 + Layer 3]

The collapsing geometry and the expanding geometry are the same structure, mapped across Σ . The difference between them is $\varepsilon \cdot h$ — a small perturbation carrying everything that makes a universe a universe. If the equation is wrong, nine kill switches will find it.

7.5 — What h encodes

Anisotropy difference. M^- is Kantowski-Sachs (anisotropic). M^+ is nearly FRW (isotropic). h carries the difference. [Layer 2]

Information content. Information is deviation from symmetry. If $\varepsilon = 0$, perfect $\mathbb{Z}_2$, no records. If $\varepsilon > 0$, h carries the records compressed into the transition. [Layer 3]

Initial conditions. The trace of h on Σ determines the initial perturbation spectrum. [Layer 2 + 3]

7.6 — The electron and α

The electron is read as the minimal low-energy carrier of the break (section 5.3), and α as the coupling strength (section 5.4). This is an interpretive identification, physically motivated but not derived from the conjecture. The value of α is not derived (D3); its underivability is consistent with AP41's structural argument.

7.7 — Limits

$\varepsilon \rightarrow 0$. $g^+ \rightarrow (\varphi^{-1})^*(g^-)$. Perfect $\mathbb{Z}_2$. No records. No distinguishable time direction. Sterile symmetry.

ε large. g^+ departs from $(\varphi^{-1})^*(g^-)$. Rich structure. But the axiom constrains ε to be small ($1 \times \varepsilon$). The perturbation is gentle.

7.8 — The full chain

Step 1. Axiom: $1:1 + 1 \times \varepsilon$. Step 2. $1:1 \equiv \mathbb{Z}_2$ on Σ (Assumptions A2, A3). Step 3. $+ 1 \times \varepsilon \equiv \mathbb{Z}_2$ breaking; $\exists \varphi: M^- \rightarrow M^+$. Step 4. Conjecture: $g^+ = (\varphi^{-1})^*(g^-) + \varepsilon \cdot h$ (Assumption A5). Step 5. h carries records; $\varepsilon \cdot h$ induces $P_m(k)$ on the expanding side. Step 6 (interpretive identification, not derived from Steps 1-5). Electron = minimal low-energy carrier of the break (section 5.3); α = coupling strength (section 5.4) — physically motivated but not derived from the conjecture. Step 7 (layer shift: entirely Layer 3). Awareness $\equiv \varepsilon > 0$ (naming choice). Structural result: all actualisation shares a common origin. Perspectival result: a conscious entity recognising this unity can identify cruelty as incoherent with it. The obligation is in the recognition, not in the structure. [Layer 3]

8 — The Structural Identity

8.1 — Established mathematics [Layer 1]

Claim L8 (Established). Schwarzschild interior = Kantowski-Sachs. For $r < 2GM$, r is timelike and t is spacelike; the interior is diffeomorphic to the Kantowski-Sachs cosmological model [Kantowski & Sachs 1966].

Claim L9 (Established). Kruskal-Szekeres extension. Region IV (past the singularity in the maximal extension) shares causal-diagram character with an FRW big bang [Kruskal 1960, Szekeres 1960].

You already know what these two claims mean together. The inside of a black hole is a collapsing universe; the maximal extension reveals an expanding universe on the other side. The mathematics has been published since 1966. The structural identity has been hiding in plain sight for sixty years.

8.2 — Speculative physics [Layer 2]

Claim L10 (Published, not established). Popławski’s mechanism. Fermion spin-torsion coupling becomes repulsive at extreme density; gravitational collapse transitions to cosmological expansion, and the resulting universe is nearly isotropic [Popławski 2010, 2016, 2023].

8.3 — Foliation independence [Layer 3]

The Einstein field equations describe a 4-dimensional manifold without a preferred global time function. The ‘bounce’ appears under specific foliations; the structural identity is atemporal. Mathematically permissible: no preferred global time function is required by the Einstein equations.

8.4 — The anisotropy problem [Layer 2]

Open Problem L-Aniso (KS-L0.8). The Schwarzschild interior is Kantowski-Sachs (anisotropic). Observation requires FRW (isotropic). Popławski proposes torsion-driven isotropisation; Ridgely [2024] argues an alternative mechanism. Neither is established. The sharpest obstacle to the structural identity.

Here is how to destroy the structural identity at its weakest point: show that no mechanism — torsion, quantum gravity, anything — can isotropise a Kantowski-Sachs interior into an FRW cosmology. If you can prove that, KS-L0.8 fires and the conjecture falls. The argument invites you to try.

8.5 — The entropy question

Consistent with the Generalised Second Law, as set out below. Maximum coarse-grained entropy: the transition hypersurface represents maximum coarse-grained entropy [Layer 1]. Unitarity: $S_{\text{vN}}(\rho_{\text{total}}) = \text{const}$; coarse-grained entropy increases; no information loss [Layer 1]. Change of description: the apparent entropy ‘reset’ is a change of coarse-graining subalgebra, not a violation of the second law [Layer 1 + 3]. Island formula: the Page curve is reproduced, S_{gen} non-decreasing [Almheiri et al. 2021] [Layer 1]. Weyl resolution: small initial Weyl tensor from scrambled correlations [Layer 2 + 3]. Multiversal GSL: Garriga et al. [2025] provide a scheme for the generalised second law in multiverse contexts [Layer 1].

8.6 — The structure experienced from inside

To an observer within the broken phase ($\varepsilon > 0$): expansion, stellar processing, gravitational compression, compression toward the quantum-gravity regime. The temporal sequence is perspectival; the underlying structure is atemporal, foliated by proper time. What is experienced as a ‘running engine’ is the atemporal structure read from inside. You are inside it now. You have always been inside it. The expansion you observe, the stars you see, the compression that awaits — all of it is the atemporal structure read from your vantage point within the broken phase.

9 — Relation to Loop Quantum Cosmology

Loop Quantum Cosmology [Ashtekar & Singh 2011] derives a bounce at the Planck density as a dynamical, temporal event. AP04 proposes the structural identity as atemporal: the bounce is foliation-dependent, the geometry atemporal, the identity structural. If LQC is experimentally confirmed, it strengthens the physical foundation (a Layer 1 candidate). The relationship is complementary, not competitive. [LQC: Layer 1 candidate. Interpretation: Layer 3.]

10 — Unique Predictions

Prediction P1 (Inherited CMB anisotropy, KS-L0.9). Residual Kantowski-Sachs patterns (quadrupolar/axial) distinct from inflationary predictions. Specific observed anomalies potentially consistent: the CMB quadrupole-octupole alignment, the hemispherical power asymmetry, the CMB cold spot. [Layer 2: testable]

Prediction P2 (No terminal singularity). Any complete quantum-gravity theory must yield an extension beyond the Planck density, not a terminal singularity. [Layer 2: falsifiable]

Prediction P3 (Fermion necessity). A bosonic-only universe should not exhibit the structural identity; the torsion mechanism requires fermionic spin. [Layer 2: testable via simulation]

Prediction P4 (Black-hole interior cosmology). Black-hole interior dynamics are cosmological, not terminating. [Layer 2: long-term]

Prediction P5 (Scale signatures). If nested (each black hole births a universe): a preferred axis, anomalous large-angle correlations, bulk flow. [Layer 2 + 3: testable, suggestive]

Prediction P6 (CMB spectrum from h , KS-L0.9). The perturbation tensor $\varepsilon \cdot h$ must induce a curvature perturbation spectrum $P(k)$ consistent with the near-scale-invariant spectrum with measured tilt $n_s \approx 0.965$. [Layer 2: quantitative, testable]

11 — The Scale Question

Cosmic. Our universe = a black-hole interior in a parent structure.

Astrophysical. Each black hole = a local site of the structural identity.

Nested. Both = the same mechanism at different scales. Fractal. [Layer 3]

12 — Relation to Existing Physics

AP04’s structural identity intersects several existing research programmes. Penrose’s Conformal Cyclic Cosmology (CCC) [Penrose 2010] proposes that the conformal boundary of one aeon becomes the big bang of the next. AP04 shares the intuition that cosmological endpoints and origins are related, but differs fundamentally: CCC is temporal (one aeon follows another), while AP04’s identity is atemporal (the structure is read from two sides of Σ , not sequenced); CCC requires conformal rescaling and massless final states, AP04 requires torsion-driven transition and fermionic matter.

Smolin’s Cosmological Natural Selection (CNS) [Smolin 1992] proposes that black holes give birth to new universes with slightly altered constants under selection pressure. AP04 shares the structural identity but does not invoke selection: the constants are determined by ε (if AP03 holds), not evolved. CNS is Darwinian; AP04 is structural. Pathria [1972] and subsequent work proposed that the observable universe might be the interior of a black hole; AP04 sharpens this to a specific conjecture (L7) with an explicit transition mechanism (Popławski’s torsion) and explicit falsification conditions. Easson and Brandenberger [2001] proposed that contraction inside a black hole could generate a new expanding phase; AP04’s contribution is the atemporal reading. Corichi and Singh [2016] showed that the loop quantum bounce preserves large-scale structure while resolving the singularity — consistent with AP04’s requirement that ε -h carry information through the transition.

13 — Kill Switches

Nine explicit conditions (canonical registry numbering KS-L0.1–KS-L0.9). If any is satisfied, the corresponding component of AP04 fails. The argument does not protect itself. It tells you where to cut.

13.1 — KS-L0.1 — Structural identity disproved

Claim. A diffeomorphism-compatible mapping exists between a black-hole interior and an expanding FRW-type cosmology under some completion of general relativity.

Test. A rigorous proof that no such mapping exists under any completion.

Status. LIVE — HARD. Targets the conjecture (L7).

Recovery. If disproved, the structural identity is dead; the established Kantowski-Sachs/Kruskal-Szekeres mathematics (L8, L9) survives as established physics without the cosmological reading.

13.2 — KS-L0.2 — No singularity resolution

Claim. A non-singular transition hypersurface Σ exists — quantum gravity extends beyond the Planck density rather than terminating.

Test. A complete quantum-gravity theory that yields a terminal singularity with no extension beyond the Planck density.

Status. LIVE — HARD. Targets the conjecture (L7).

Recovery. A terminal singularity removes Σ and collapses the conjecture; the symmetry-break reading of ε would have to be relocated.

13.3 — KS-L0.3 — Information loss

Claim. Black-hole evaporation is unitary; information is scrambled, not destroyed (the entropy resolution of section 8.5).

Test. A demonstration that evaporation is fundamentally non-unitary.

Status. LIVE — EMPIRICAL. Targets section 8.5.

Recovery. Non-unitarity breaks the entropy resolution; Axiom R (records persist) would be in direct tension and require re-examination.

13.4 — KS-L0.4 — Proton decay observed

Claim. The fermion-dependent mechanism is consistent with the observed stability of the proton.

Test. Proton decay observed on timescales relevant to the structural identity (indicating baryon-number violation inconsistent with the mechanism).

Status. LIVE — EMPIRICAL. Targets the fermion-necessity claim (P3, L10).

Recovery. Observed proton decay weakens the fermion-necessity claim without immediately collapsing the structural identity.

13.5 — KS-L0.5 — GSL violated

Claim. The Generalised Second Law holds across the transition ($\Delta S_{\text{gen}} \geq 0$).

Test. A demonstration of $\Delta S_{\text{gen}} < 0$ in a closed region.

Status. LIVE — EMPIRICAL. Targets section 8.5.

Recovery. A GSL violation falsifies the thermodynamic resolution; the structural identity would need an alternative entropy account.

13.6 — KS-L0.6 — Actualisation without awareness

Claim. Actualisation (record formation) does not occur without relational structure (the awareness identification, L6).

Test. Experimental confirmation of objective collapse (GRW) — states becoming actual without relational structure.

Status. LIVE — EMPIRICAL. Targets L6.

Recovery. Confirmed GRW kills the identification of ε with awareness; the physics of the conjecture is unaffected (the naming changes, the structure does not).

13.7 — KS-L0.7 — Fermions unnecessary

Claim. Fermionic spin-torsion is necessary to the transition mechanism (Popławski).

Test. A bosonic-only universe shown to reproduce the structural identity without fermionic torsion.

Status. LIVE — HARD. Targets L10, P3.

Recovery. If fermions are unnecessary, Popławski's mechanism is not required and the isotropisation route (D1) must be re-grounded.

13.8 — KS-L0.8 — Anisotropy irreconcilable

Claim. Some mechanism isotropises the Kantowski-Sachs interior to FRW-like isotropy.

Test. A proof that no mechanism can isotropise the Kantowski-Sachs interior to FRW.

Status. LIVE — HARD. The sharpest obstacle (D1).

Recovery. If isotropisation is impossible, the structural identity cannot connect black-hole interiors to observed cosmology; the conjecture falls at its weakest point.

13.9 — KS-L0.9 — CMB mismatch

Claim. The perturbation tensor $\varepsilon \cdot h$ reproduces the observed curvature perturbation spectrum $P(k)$ (near-scale-invariant, $n_s \approx 0.965$), and inherited Kantowski-Sachs patterns are consistent with the observed CMB anomalies (P1, P6).

Test. If $\varepsilon \cdot h$ cannot reproduce $P(k)$ under any reasonable parameterisation of h , the conjecture makes wrong predictions.

Status. LIVE — EMPIRICAL. Targets the conjecture via P1, P6 (D2).

Recovery. A CMB mismatch falsifies the predictive content; the structural identity could survive as an interpretation without the specific spectrum prediction, pending the derivation of $P_m(k)$ from $\varepsilon \cdot h$ (D2).

14 — What This Establishes and What Remains Open

14.1 — What this paper establishes

It establishes (Layer 1) the mathematics of the structural identity: the Schwarzschild interior as a Kantowski-Sachs cosmology (L8), the Kruskal-Szekeres extension (L9), the two constraints c and ρ_p (L1, L2), and the entropy resolution consistent with the Generalised Second Law and the Page curve (section 8.5). It formalises the axiom as a conjecture (L7) under five explicit assumptions and lays out nine falsification conditions.

14.2 — What this paper does not close

The conjecture L7 is not derived — it is held at Layer 2/3 and guarded by KS-L0.1, KS-L0.2, KS-L0.8, KS-L0.9. The isotropisation mechanism (D1, KS-L0.8) is the sharpest open obstacle; the CMB spectrum derivation (D2, KS-L0.9) is open; the value of α (D3) is not derived.

14.3 — Items held open without claiming debt status

The awareness identification (L6) is a Layer 3 naming choice, falsifiable only via GRW (KS-L0.6). The scale question (cosmic / astrophysical / nested) is a Layer 3 reading. The interpretive identification of the electron and α (Step 6) is physically motivated but not derived from the conjecture.

14.4 — Where the philosophical-register implementations live

The expository-register companion for the loop, the awareness identification, and the ethics lives in the $\emptyset$ Models and in AP41 (the body of work-level statement of the loop and the dark sector). AP04 is the formal-register origin of the structural-identity conjecture.

14.5 — Structural relationships to subsequent work

AP41 (The Loop) inherits AP04's conjecture at Stage 6 (KS-41.3 inherits KS-L0.1) and provides the body of work-level statement of the underivability of α (D3). AP03 supplies the conjugacy of the two constraints.

14.6 — The 420 Code placement

AP04 is a chapter of Notebook VI (Cosmology) — the loop hypothesis, the conjectural foundation that the dark-sector and loop chapters (AP41, AP42) build on and guard.

14.7 — How the body of work reads now

With AP04 in place, the 420 Code carries an explicit, falsifiable conjecture for the loop — a metric equation across a transition hypersurface, layered by certainty, with nine kill switches and the sharpest obstacle (isotropisation) named openly. The downstream cosmology (AP41, AP42) treats the loop-closure step as conjectural and guarded, exactly as AP04 holds it.

14.8 — The closing

The inside of a black hole is a collapsing universe; the maximal extension reveals an expanding universe on the other side. The two are the same structure, mapped across a transition hypersurface, differing by $\varepsilon \cdot h$ — a small perturbation carrying everything that makes a universe a universe. This is a conjecture, not a derivation, and it tells you where to cut: prove that no mechanism can isotropise the Kantowski-Sachs interior into FRW, and it falls. You are inside the broken phase now. The expansion you observe, the stars you see, the compression that awaits — all of it is the atemporal structure read from your vantage point within the break.

Claim Summary

ESTABLISHED (Layer 1). L1 (c , the null cone), L2 (the Planck density $\rho_p = c^5/(\hbar G^2) \approx 5.16 \times 10^{96} \text{ kg/m}^3$), L8 (Schwarzschild interior = Kantowski-Sachs), L9 (the Kruskal-Szekeres extension), and the entropy resolution (section 8.5: unitarity, the island formula, the GSL).

CALCULATION / ESTABLISHED VALUE (Layer 1 + 3). L3 (the inverse relationship; necessary but not sufficient), L4 (the electron violates Kerr-Newman by $\sim 10^{89}$, spin-dominated), L5 ($\alpha = e^2/(4\pi\epsilon_0\hbar c) \approx 1/137$, value not derived).

PUBLISHED, NOT ESTABLISHED (Layer 2). L10 (Popławski's torsion mechanism), the predictions P1-P6.

CONJECTURE (Layer 2 + 3). L7: $g^+ = (\varphi^{-1})^*(g^-) + \varepsilon \cdot h$, under Assumptions A1-A5, with Interpretations I1 ($1:1 \equiv \mathbb{Z}_2$ on Σ) and I2 ($+1 \times \varepsilon \equiv \mathbb{Z}_2$ breaking). Not derived; guarded by KS-L0.1, KS-L0.2, KS-L0.8, KS-L0.9.

PHILOSOPHICAL (Layer 3). L6 (awareness $\equiv \varepsilon > 0$; the two-part ethics result), the foliation independence, the structure-from-inside reading, the scale question.

OPEN DEBT. D1 (isotropisation, KS-L0.8 — the sharpest obstacle), D2 (the CMB spectrum from $\varepsilon \cdot h$, KS-L0.9), D3 (the value of α — consistent with AP41's structural underivability argument).

FALSIFICATION REGISTER. KS-L0.1 (structural identity, HARD), KS-L0.2 (singularity resolution, HARD), KS-L0.3 (information loss, EMPIRICAL), KS-L0.4 (proton decay, EMPIRICAL), KS-L0.5 (GSL, EMPIRICAL), KS-L0.6 (GRW / awareness, EMPIRICAL), KS-L0.7 (fermions, HARD), KS-L0.8 (anisotropy, HARD — sharpest obstacle), KS-L0.9 (CMB, EMPIRICAL).

Conditionality Footer

Dependencies

Established relativity (Kantowski & Sachs 1966; Kruskal-Szekeres; Bekenstein-Hawking; Almheiri et al. 2021), published Einstein-Cartan torsion (Popławski 2010/2016/2023), and within the body of work AP03 (the conjugacy of c and ρ_p) and AP20 (EH/QRA proved).

Dependents

AP41 (inherits the conjecture at Stage 6; KS-41.3 inherits KS-L0.1) and the downstream Notebook VI cosmology that treats the loop-closure step as conjectural and guarded.

Kill switches closed by this paper

None. KS-L0.1 through KS-L0.9 are introduced and remain live (four HARD: KS-L0.1, KS-L0.2, KS-L0.7, KS-L0.8; five EMPIRICAL: KS-L0.3, KS-L0.4, KS-L0.5, KS-L0.6, KS-L0.9).

Kill switches that remain live

All nine: KS-L0.1 through KS-L0.9.

Structural debts owed

D1 (isotropisation, KS-L0.8), D2 (CMB spectrum, KS-L0.9), D3 (the value of α ; consistent with AP41's structural underivability argument). All open.

Items held open without claiming debt status

The awareness identification (L6, Layer 3, falsifiable via KS-L0.6); the scale question (Layer 3); the interpretive identification of the electron and α (Step 6, not derived from the conjecture).

Notation Reference

1:1 + 1× ε — Axiom L0: one-to-one symmetry broken by an infinitesimal perturbation $\varepsilon > 0$.

ε — the symmetry-breaking parameter; the break. Also identified (Layer 3) with awareness.

$\mathbf{g}^+, \mathbf{g}^-$ — the metrics on the expanding side (M^+) and the collapsing side (M^-) of the transition.

φ, Σ — the transition diffeomorphism $\varphi: M^- \rightarrow M^+$ ($\varphi|_\Sigma = \text{id}$, normal-reversing) and the non-singular transition hypersurface Σ .

$(\varphi^{-1})^*$ — the pullback of φ^{-1} , mapping tensors on M^- to tensors on M^+ .

$\mathbf{h}$ — the perturbation tensor (symmetric 2-tensor) encoding the anisotropy difference, the information content, and the initial conditions; $\varepsilon \cdot \mathbf{h}$ is the deviation across the transition.

$\mathbb{Z}_2$ — the two-element symmetry group; $1:1 \equiv \mathbb{Z}_2$ on Σ (I1), $+1 \times \varepsilon \equiv \mathbb{Z}_2$ breaking (I2).

$\mathbf{c}, \rho_p$ — the propagation constraint (the speed of light; the null cone) and the compression constraint (the Planck density, $\rho_p = c^5/(\hbar G^2)$).

ℓ_p, t_p — the Planck length and Planck time; $\ell_p/t_p = c$.

α — the fine-structure constant, $\alpha = e^2/(4\pi\epsilon_0\hbar c) \approx 1/137$; value not derived (D3).

$\mathbf{P}_m(\mathbf{k}), \mathbf{n}_s$ — the curvature perturbation spectrum and its measured tilt ($\mathbf{n}_s \approx 0.965$).

S_{vN}, S_{gen} — the von Neumann (fine-grained) entropy and the generalised entropy; $S_{vN}(\rho_{total}) = \text{const}$, S_{gen} non-decreasing.

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